



Magnetic properties of amorphous metallic composites with various particle sizes

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Abstract

Recently, soft magnetic composites (SMCs) have been investigated considerably because of an incessant need for high performance in the high-frequency range. If high performance is to be achieved at high frequencies, the permeability must be increased, and core-loss reduced. In this study, we examined the magnetic properties and the core loss of a soft magnetic composite based on a Fe amorphous alloy with different particle sizes. The amorphous Fe alloy powders with different average sizes from 25 to 63 μm were prepared as raw materials using sieving. Powders were prepared using a gas atomizing process. The DC magnetic properties of the synthesized cores were studied with a vibrating sample magnetometer (VSM), and the permeability and the core-loss measurements were conducted on an impedance analyzer and a B – H analyzer, respectively.

Keywords Soft magnetic composites · Permeability · Amorphous

1 Introduction

Fe-based soft magnetic alloys are key functional materials widely used in electric devices and communication fields [1–3]. However, most Fe-based magnetic alloys exhibit poor magnetic characteristics in the high-frequency range, so they are used only in the low-frequency range. For high-frequency application, soft magnetic composites have attracted scientific interest because these materials exhibit viable overall performance with high magnetic induction, low core loss, and high working frequency, in latent demand for passive components such as power inductors and motors [4, 5]. Amorphous powder has viable DC magnetic properties, such as high saturation magnetization, low coercivity, and high permeability [7, 8]. For this reason, the composite consisting of an amorphous metal powder-insulating polymer composite is a viable candidate for electric devices with high magnetic properties [9]. The amorphous Fe alloy is

one of the base materials for applications working in the high-frequency range because of the low eddy current loss induced by the high specific resistance [10, 11]. The microstructure of the core is another factor affecting the magnetic properties of the core. When core packing density is optimized, the magnetic performance is highly increased, and the optimized microstructure can be attained by selecting a powder with an appropriate average particle size [12, 13].

In this study, we examined the magnetic properties of soft magnetic materials (SMCs) with different average particle size. We will discuss the magnetic properties of an amorphous-resin composite, and their dependencies on the size of the particles in each amorphous powder.

2 Experimentals

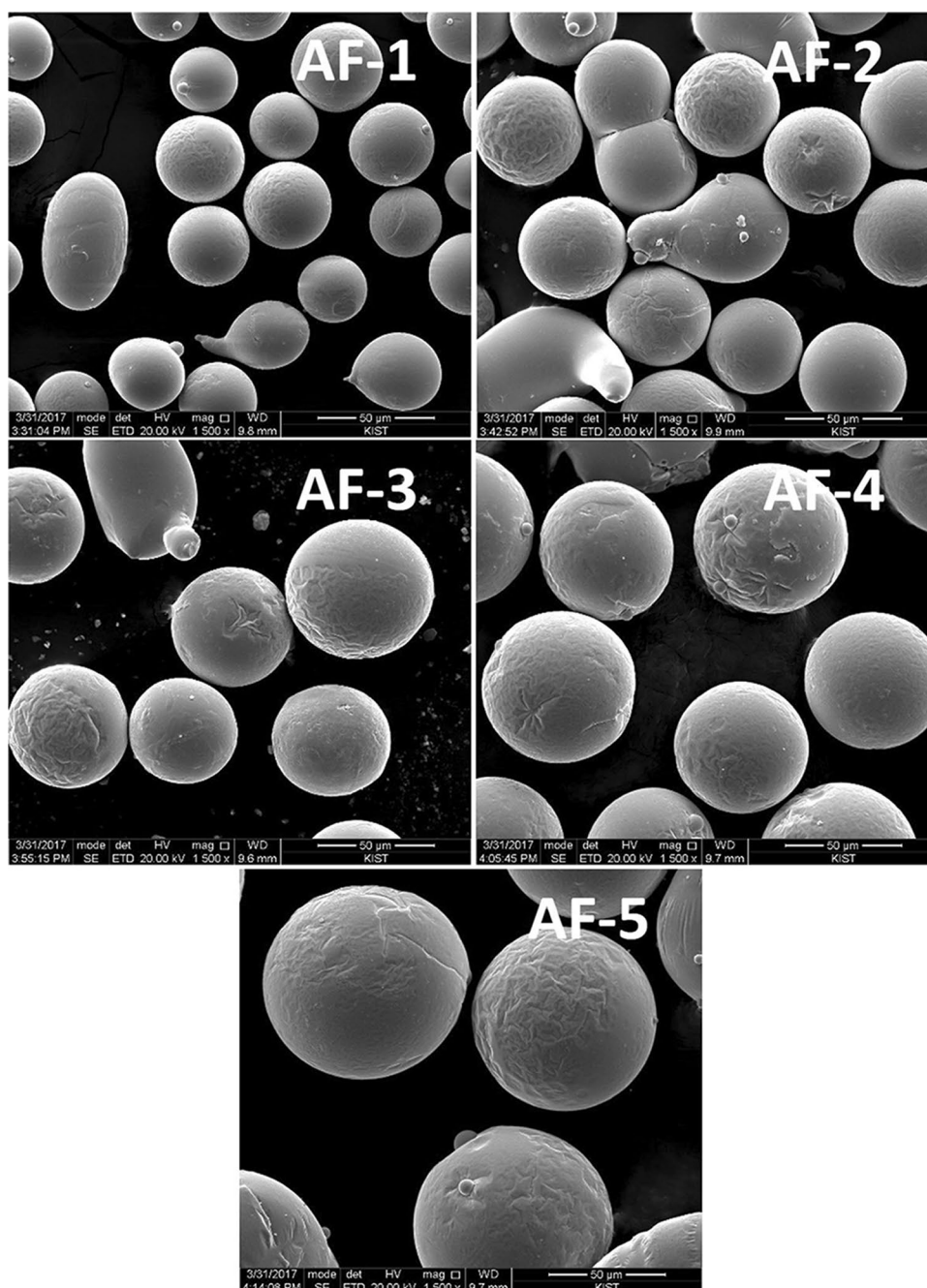
An amorphous Fe alloy powder was developed using the gas atomizing method. The composition of the synthesized powder was $\text{Fe}_{86.5}\text{Si}_{4.2}\text{Nb}_{3.4}\text{B}_{1.7}\text{P}_{1.5}\text{Mo}_{1.4}\text{C}_{1.3}$ (wt%), and the initial sizes of the particles in each powder ranging from 25 to 200 μm . The amorphous Fe alloy powders were classified into five size ranges using sieves with each mesh size ranging from 25 to 63 μm . The sizes of particles in each powder were categorized as 25–38 μm , 38–45 μm , 45–53 μm , 53–63 μm , and over 63 μm . The sieved powder in each range

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Fig. 1 Morphologies of the sieved amorphous Fe alloy powders



were coded as follows: 25–38 μm to ‘AF1’, 38–45 μm to ‘AF2’, 45–53 μm to ‘AF3’, 53–63 μm to ‘AF4’, and over 63 μm to ‘AF5’, respectively. For forming a mixing sample toroidal core, blending with a 1.5 wt% epoxy binder, we fabricated the composite core by pressing at pressure of 1.5 ton/cm^2 at room temperature and using a heat treatment at 160 $^{\circ}\text{C}$ for 2 h in ambient air to cure of the core. Each core synthesized using amorphous powder sieved in the five ranges were coded as follows: 25–38 μm to ‘AF1-C’, 38–45 μm to ‘AF2-C’, 45–53 μm to ‘AF3-C’, 53–63 μm to ‘AF4-C’, and over 63 μm to ‘AF5-C’, respectively. The

characteristics of the amorphous and the Fe powder were analyzed using X-ray diffraction (XRD) with $\text{Cu } \alpha$, in a range of 2 theta = 20–90 $^{\circ}$, in steps of 0.01 $^{\circ}$, vibration sample magnetometry (VSM) and scanning electron microscopy (SEM). The magnetic properties of toroidal cores were measured using an impedance analyzer to obtain the permeability, quality factor, and frequency-dependent specificity. Core loss was measured with a B – H analyzer at various frequencies and applied fields.

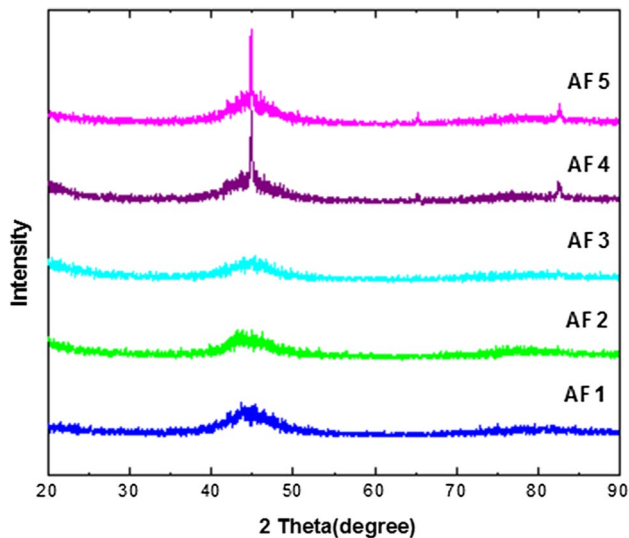


Fig. 2 X-ray diffraction patterns of the sieved amorphous Fe alloy powders

3 Results and discussion

Figure 1 shows amorphous powder morphology for each powder. The size distribution of each powder was seen to be similar with the mesh size used in sieving the powders. The particles in the AF1 powder were almost spherical with the diameter differences being affected by solidification rate. Powders with larger particle sizes had lower solidification rate, due to the slow heat transfer rate whereas powders with smaller particle size had higher solidification rates [14]. The particles in the amorphous powders have smooth surfaces with less contamination and fewer adhering satellites.

X-ray diffraction patterns are shown in Fig. 2. Observed peaks of amorphous powders below AF3 reveal considerably weak intensities and broad shapes. Such weak and broad peaks are typical characteristics of an amorphous structure. The patterns of both AF4 and 5 revealed an obvious first-order peak at approximately $2\theta = 44.82^\circ$, indicating that AF4 and 5 have mixed phases, an amorphous phase mixed with crystalline α -Fe. Results suggest the amorphous phase is formed due to rapid cooling in a powder with small particles whereas in the case of powders with large particles, crystal phase is mixed due to cooling rate being slow compared to that of powders with small particle sizes.

In Fig. 3, the field dependences of the magnetizations for all amorphous Fe alloy powders are presented. The amorphous AF1, AF2, and AF3 powders revealed similar saturation magnetizations at approximately 131 emu/g, with the saturation magnetizations of AF4 and AF5 (~ 130 emu/g) being slightly lower than those of AF1 to AF3. However, in the case of the coercivity, AF 4 and AF5 showed obviously higher coercivities (over 1.2 Oe) than did AF1 to AF3

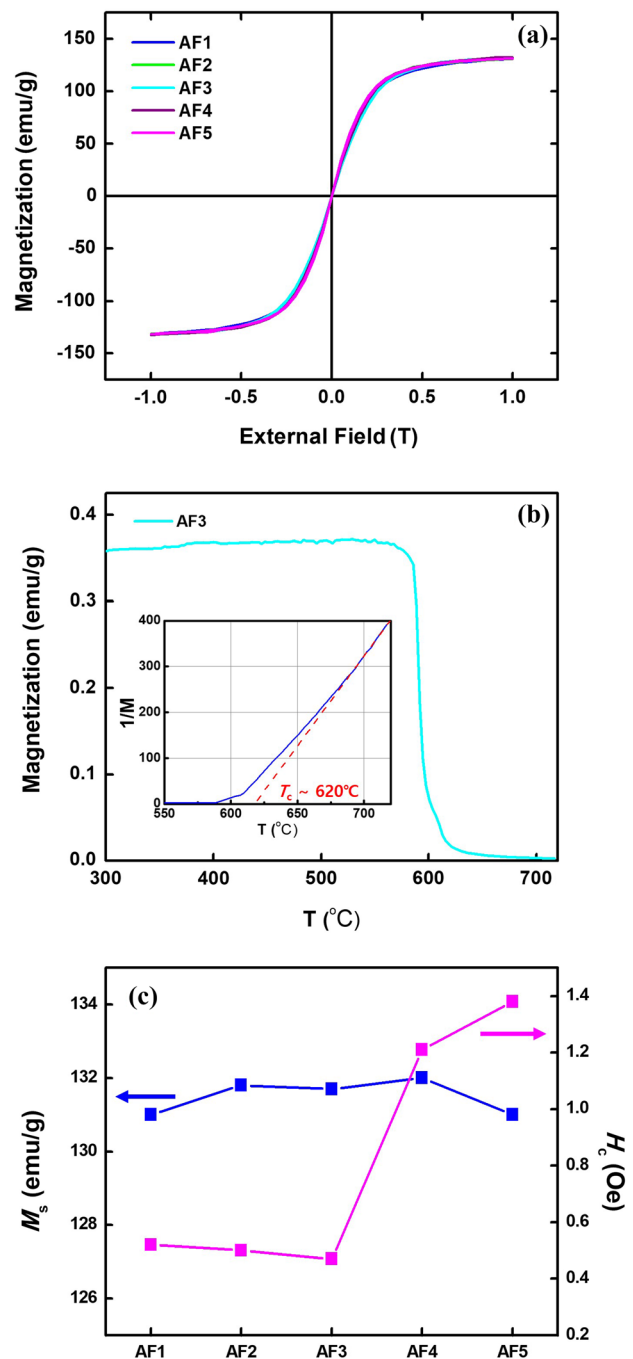


Fig. 3 Magnetic hysteresis properties of the sieved amorphous Fe alloy powders

(around 0.5 Oe). These coercivity increases were caused by the α -Fe crystalline phase in AF4 and AF5. Crystalline Fe alloy powders were reported to have high coercivities compared with amorphous Fe alloy powders [13]. The above crystalline analysis revealed that amorphous Fe alloy powders with average particle sizes below $53\ \mu\text{m}$ were in a

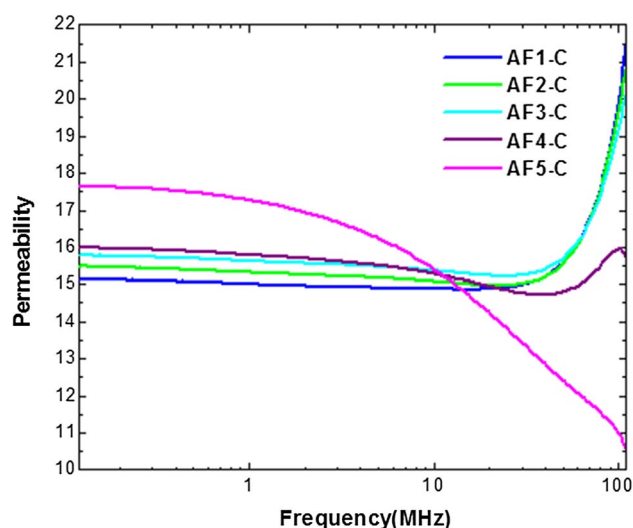


Fig. 4 Frequency dependence of μ swept in frequency range from 1 kHz to 110 MHz

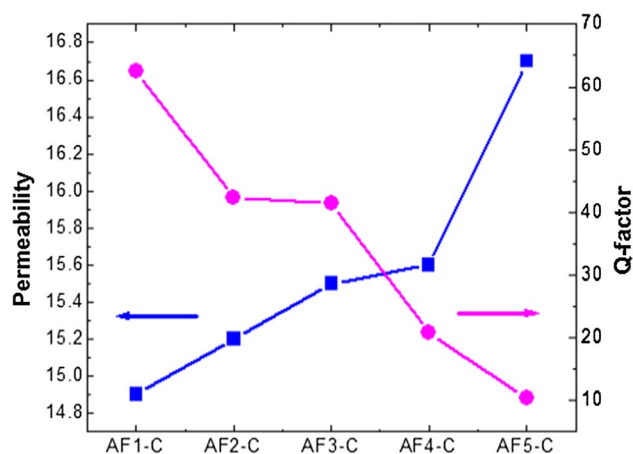


Fig. 5 The μ and the Q factor at 1 MHz for the toroidal cores according to constituent particle size

pure amorphous phase. The results from the DC magnetic properties evaluation, especially for AF 1 to AF3, revealed low coercivities below 1 Oe, strongly confirming the above results obtained from SEM and XRD. The M - T curve of AF3 (Fig. 3b), measured at an applied field 200 Oe, revealed a Curie temperature of approximately 620 °C without any secondary phase. It also indicated the presence of an amorphous powder with high purity.

The frequency dependences of relative permeabilities of the synthesized toroidal cores were measured in the frequency range from 1 kHz up to 110 MHz (Fig. 4). As the particle size increased, the permeability of the composite increased gradually. Because the permeability is strongly influenced by the size of the particles in an amorphous powder, the permeabilities of toroidal cores from AF1-C

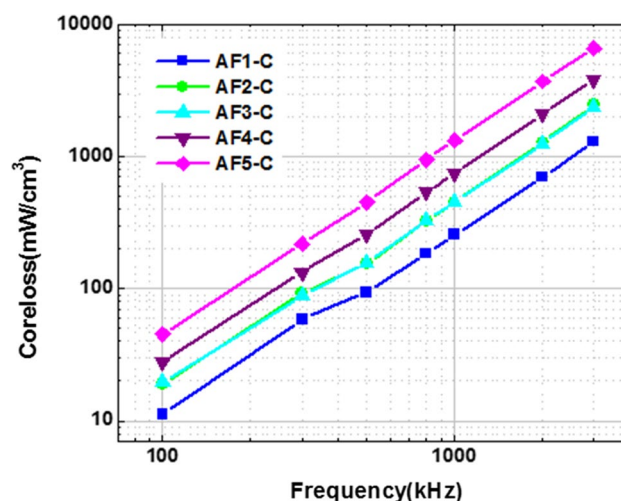


Fig. 6 Results from a core-loss swept in the frequency range from 100 to 3000 kHz

Table 1 The observed magnetic parameters of cores

Sample	M_s (emu/g)	μ @ 3 MHz	Q @ 3 MHz	Core loss (mW/cm ³)
AF1-C	130.9	14.9	62.47	1290
AF2-C	131.7	15.2	42.38	2446
AF3-C	131.6	15.5	41.48	2367
AF4-C	131.9	15.6	20.84	3816
AF5-C	131.0	16.7	10.46	6604

to AF 3-C were stable up to 30 MHz, but that of AF4-C dropped sharply at 100 MHz. The permeability of AF5-C was seen to be higher than that of other cores at frequencies below 10 MHz, and to decrease at approximately 2 MHz. From the frequency characteristics of the composite core, amorphous Fe alloys with powder containing particles up to 53 μ m in size had superior AC magnetic behaviors in the high-frequency range.

The permeabilities and the Q factors measured at 3 MHz of the synthesized toroidal cores are shown in Fig. 5. As the size of the particles in the amorphous powder of the toroidal cores becoming larger, the magnetic permeability was gradually enhanced from 14.8 for AF1-C to 16.7 for AF5-C. Especially, the permeability variation between AF4-C and AF5-C was higher than it was for the others. The high permeability observed for AF5-C was the result of the presence of particles with sizes of several hundred μ m in the powder used for the composite core. Observations suggest the permeability of a toroidal core has a dependency on the constituent particle size and this dependency is induced by an interaction between magnetic particles and the domain structure [15–17]. The Q factor decreases as particle size

increases, unlike the permeability characteristics of the composite core. The Q factor of AF1-C core was highest value at 62 and decreased gradually as constituent powder size increased. The Q characteristics of the AF2-C and the AF3-C cores were 42.38 and 41.48, respectively, whereas the Q of AF4-C sharply decreased to less than 20, which were the result of mixing a highly crystalline powder.

In Fig. 6, the frequency dependence of the core loss according to constituent powder size is presented. The core loss of the AF1-C core was measured as 1290 mW/cm³ at 3 MHz for $B_m = 10$ mT. The core loss is seen to increase linearly as constituent powder size rises at every step. The increase in the core loss of powders with larger particles causes an increase in the eddy current loss induced by intra-particle eddy currents.

In highly resistive regions in which insulation between particles is maintained, eddy currents are conducted only inside the powder without passing through the powder next to the powder. When the skin depth is deeper than the core thickness, eddy current loss is inversely proportional to the resistance and proportional to the square of the thickness. For the above reasons, the core loss shown in Fig. 6 indicates that the eddy current loss increases as the particle size in the powder of the composite increases.

The magnetic properties and core loss are shown in Table 1. The permeability is directly proportional to the particle size; conversely, Q is inverse proportional. Core loss also increases with increasing constituent particle size. The permeability is affected by the particle size, whereas the Q value and the core loss are affected by eddy currents inside the powder. Particularly, the variations in the Q value and the core loss from AF4-C to AF5-C were caused by the crystallinity of the mixed powder.

4 Conclusion

In this study, the permeability, Q factor, and core loss dependency as functions of the constituent particle size were investigated. The permeability was found to change due to the difference in the size of the particles in the powder, and the core loss was found to vary with the particle size and the crystal structure. To fabricate a soft magnetic composite with high magnetic performance in the high-frequency

range, an amorphous powder with an appropriate powder size must be used to ensure a magnetic permeability and a core loss suitable for the usage condition. From the results, we conclude that a core composed of powders with particles with sizes of a few tens of micrometer revealed excellent and stable AC magnetic properties and low loss.

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